1. Demostrar que para una red ortorombica $a \neq b \neq c$, $\alpha = \beta = \gamma = 90°$; Se cumple la relación

$$|\vec{G}_{hkl}| = \frac{2\pi}{d_{hkl}}$$

Demostración:

$$\vec{a}_1 = a\hat{i}, \quad \vec{a}_2 = b\hat{j}; \quad \vec{a}_3 = c\hat{k}$$

Espacio reciproco $\{\vec{b}_1, \vec{b}_2, \vec{b}_3\}$

1
3 2

$$\vec{b}_1 = \frac{2\pi\, \vec{a}_2 \times \vec{a}_3}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)}; \quad \vec{b}_2 = \frac{2\pi\, \vec{a}_3 \times \vec{a}_1}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)}$$

$$\vec{b}_3 = \frac{2\pi\, \vec{a}_1 \times \vec{a}_2}{\vec{a}_1 \cdot (a_2 \times \vec{a}_3)}$$

$$\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)$$

2

$$\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3) = \begin{vmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{vmatrix} = abc \checkmark$$

$$\vec{a}_2 \times \vec{a}_3 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & b & 0 \\ 0 & a & c \end{vmatrix} = bc\,\hat{i}$$

$$\vec{a}_3 \times \vec{a}_1 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & c \\ a & 0 & 0 \end{vmatrix} = ac\,\hat{j}$$

$$\vec{a}_1 \times \vec{a}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a & 0 & 0 \\ 0 & b & 0 \end{vmatrix} = ab\,\hat{k}$$

$\vec{b}_1 = \dfrac{2\pi \cancel{bc}\,\hat{i}}{a\cancel{bc}}$ ; $\vec{b}_1 = \dfrac{2\pi}{a}\hat{i}$ ; $\vec{b}_2 = \dfrac{2\pi \cancel{ac}\,\hat{j}}{\cancel{a}b\cancel{c}}$

$\vec{b}_2 = \dfrac{2\pi}{b}\hat{j}$ ; $\vec{b}_3 = \dfrac{2\pi \cancel{ab}\,\hat{k}}{\cancel{ab}c} = \dfrac{2\pi}{c}\hat{k}$

$\vec{b}_1 = \dfrac{2\pi}{a}\hat{i}$ ; $\vec{b}_2 = \dfrac{2\pi}{b}\hat{j}$ ; $\vec{b}_3 = \dfrac{2\pi}{c}\hat{k}$

$\vec{b}_1 \cdot \vec{b}_2 = \vec{b}_2 \cdot \vec{b}_3 = 0$ ; $|\vec{b}_1| = \dfrac{2\pi}{a}$ ; $|\vec{b}_2| = \dfrac{2\pi}{b}$

$|\vec{b}_3| = \dfrac{2\pi}{c}$

3

$$\vec{G}_{hkl} = h\vec{b}_1 + k\vec{b}_2 + l\vec{b}_3$$

$$\vec{G}_{hkl} = h\frac{2\pi}{a}\hat{i} + k\frac{2\pi}{b}\hat{j} + l\frac{2\pi}{c}\hat{k}$$

$$|\vec{G}_{hkl}| = \sqrt{h^2\left(\frac{2\pi}{a}\right)^2 + k^2\left(\frac{2\pi}{b}\right)^2 + l^2\left(\frac{2\pi}{c}\right)^2}$$

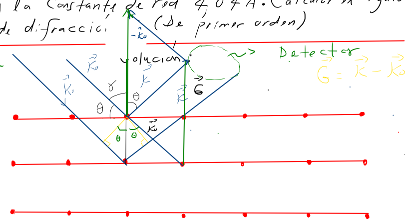
$$|\vec{G}_{hkl}| = 2\pi\boxed{\sqrt{\frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}}}$$

$$\boxed{|\vec{G}_{hkl}| = \frac{2\pi}{d_{hkl}}}$$

② En un experimento de difracción de rayos X sobre aluminio, se genera luz dispersada por la ley de Bragg desde los planos (111) utilizando una longitud de onda correspondiente a una energía de 8 KeV. La estructura del aluminio es FCC y la constante de red 4,04 Å. Calcular el ángulo de difracción (De primer orden)

Solución

1) Diagrama




2) Condición de difracción $\lambda \leq a$; $a = 4{,}04\,Å$

$E = \frac{hc}{\lambda}$ ∴ $\lambda = \frac{1240\, eV \cdot nm}{E\,(eV)}$

$$\lambda = \frac{1240\, \cancel{eV} \cdot nm}{8 \times 10^3\, \cancel{eV}} = \frac{1240\ nm}{8 \times 10^3} =$$

$\lambda = 0{,}15\ nm = 1{,}5\ Å$; $\lambda < a$; Se cumple

La condición de difracción

3) Ángulo de difracción $\boxed{2 d_{hkl} \operatorname{Sen} \theta_{hkl} = n\lambda}$

$d_{hkl}$: ?
$n$: ?
$\lambda$: ?

puedo calcular $\theta_{hkl}$ ✓

$n$: orden de difracción (número entero)

$n = 1$;

$\lambda = 1{,}5\ Å$;

$d_{hkl}$: ?

$$|\vec{G}_{hkl}| = \frac{2\pi}{d_{hkl}} \therefore$$

$$d_{hkl} = \frac{2\pi}{|\vec{G}_{hkl}|};$$

Estructura FCC:

$\vec{a_1} = \frac{a}{2}(\hat{i} + \hat{j})$; $\vec{a_2} = \frac{a}{2}(\hat{j} + \hat{k})$; $\vec{a_3} = \frac{a}{2}(\hat{i} + \hat{k})$

$\vec{G}_{hkl} = h\vec{b_1} + k\vec{b_2} + l\vec{b_3}$; ; 3 ' 2)

$$\vec{b_1} = \frac{2\pi\ \vec{a_2} \times \vec{a_3}}{\vec{a_1} \cdot (\vec{a_2} \times \vec{a_3})}; \quad \vec{b_2} = \frac{2\pi\ \vec{a_3} \times \vec{a_1}}{\vec{a_1} \cdot (\vec{a_2} \times \vec{a_3})}; \quad \vec{b_3} = \frac{2\pi\ \vec{a_1} \times \vec{a_2}}{\vec{a_1} \cdot (\vec{a_2} \times \vec{a_3})}$$

$$\vec{a_1} \cdot (\vec{a_2} \times \vec{a_3}) = \begin{vmatrix} a/2 & a/2 & 0 \\ 0 & a/2 & a/2 \\ a/2 & 0 & a/2 \end{vmatrix} = \frac{a}{2}\left(\frac{a^2}{4}\right) + \frac{a}{2}\left(\frac{a^2}{4}\right)$$

$$\boxed{\vec{a_1} \cdot (\vec{a_2} \times \vec{a_3}) = \frac{a^3}{4}} \quad ①$$

$$\vec{a_2} \times \vec{a_3} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & a/2 & a/2 \\ a/2 & 0 & a/2 \end{vmatrix} = \frac{a^2}{4}\hat{i} + \frac{a^2}{4}\hat{j} - \frac{a^2}{4}\hat{k}$$

$$\boxed{\vec{b_1} = \frac{2\pi}{a}\left(\hat{i} + \hat{j} - \hat{k}\right)}$$

$$\vec{a_3} \times \vec{a_1} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a/2 & 0 & a/2 \\ a/2 & a/2 & 0 \end{vmatrix} = -\frac{a^2}{4}\hat{i} + \frac{a^2}{4}\hat{j} + \frac{a^2}{4}\hat{k}$$

$$\boxed{\vec{b_2} = \frac{2\pi}{a}\left(-\hat{i} + \hat{j} + \hat{k}\right)}$$

$$\vec{a_1} \times \vec{a_2} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a/2 & a/2 & 0 \\ 0 & a/2 & a/2 \end{vmatrix} = \frac{a^2}{4}\hat{i} - \frac{a^2}{4}\hat{j} + \frac{a^2}{4}\hat{k}$$

$$\vec{b_3} = \frac{2\pi}{a}\left(\hat{i} - \hat{j} + \hat{k}\right)$$

$$\vec{G}_{hkl} = h\vec{b_1} + k\vec{b_2} + l\vec{b_3}$$

③

$$|\vec{G}_{hkl}|^2 = \vec{G}_{hkl} \cdot \vec{G}_{hkl}$$

$$|\vec{G}_{hkl}|^2 = (h\vec{b}_1 + k\vec{b}_2 + l\vec{b}_3) \cdot (h\vec{b}_1 + k\vec{b}_2 + l\vec{b}_3)$$

$$= h^2 \vec{b}_1 \cdot \vec{b}_1 + hk\, \vec{b}_1 \cdot \vec{b}_2 + hl\, \vec{b}_1 \cdot \vec{b}_3 + kh\, \vec{b}_2 \cdot \vec{b}_1$$
$$+ k^2 \vec{b}_2 \cdot \vec{b}_2 + kl\, \vec{b}_2 \cdot \vec{b}_3 + lh\, \vec{b}_3 \cdot \vec{b}_1 + lk\, \vec{b}_3 \cdot \vec{b}_2$$
$$+ l^2 \vec{b}_3 \cdot \vec{b}_3$$

$$= h^2 |\vec{b}_1|^2 + k^2 |\vec{b}_2|^2 + l^2 |\vec{b}_3|^2 + 2hk\, \vec{b}_1 \cdot \vec{b}_2$$
$$+ 2hl\, \vec{b}_1 \cdot \vec{b}_3 + 2kl\, \vec{b}_2 \cdot \vec{b}_3$$

Los planos difractados son $(hkl) = (1\,1\,1)$

$$|\vec{G}_{111}|^2 = |\vec{b}_1|^2 + |\vec{b}_2|^2 + |\vec{b}_3|^2 + 2\vec{b}_1 \cdot \vec{b}_2 + 2\vec{b}_1 \cdot \vec{b}_3$$
$$+ 2\vec{b}_2 \cdot \vec{b}_3$$

$$\vec{b}_1 \cdot \vec{b}_2 = \frac{2\pi}{a}(\hat{i} + \hat{j} - \hat{k}) \cdot \frac{2\pi}{a}(-\hat{i} + \hat{j} + \hat{k})$$

$$\vec{b}_1 \cdot \vec{b}_2 = -\frac{4\pi^2}{a^2}$$

$$\vec{b}_1 \cdot \vec{b}_3 = \frac{2\pi}{a}(\hat{i} + \hat{j} - \hat{k}) \cdot \frac{2\pi}{a}(\hat{i} - \hat{j} + \hat{k})$$

$$\vec{b}_1 \cdot \vec{b}_3 = -\frac{4\pi^2}{a^2}$$

$$\vec{b}_2 \cdot \vec{b}_3 = \frac{2\pi}{a}(-\hat{i} + \hat{j} + \hat{k}) \cdot \frac{2\pi}{a}(\hat{i} - \hat{j} + \hat{k})$$
$$= -\frac{4\pi^2}{a^2}$$

$$|\vec{b}_1|^2 = \frac{12\pi^2}{a^2} = |\vec{b}_2|^2 = |\vec{b}_3|^2$$

$$|\vec{G}_{111}|^2 = \frac{12\pi^2}{a^2} + \frac{12\pi^2}{a^2} + \frac{12\pi^2}{a^2} - \frac{8\pi^2}{a^2} - \frac{8\pi^2}{a^2} - \frac{8\pi^2}{a^2}$$

$$= \frac{12\pi^2}{a^2}; \quad |\vec{G}_{111}| = \frac{\pi 2\sqrt{3}}{a}$$

$$d_{111} = \frac{2\pi}{|\vec{G}_{111}|} = \frac{\cancel{2\pi}}{\frac{\cancel{\pi 2}\sqrt{3}}{a}} = \frac{a}{\sqrt{3}} = \frac{a\sqrt{3}}{3}$$

$$2d_{111} \operatorname{Sen}\theta_{111} = 1{,}5\ \text{Å}$$

$$\operatorname{Sen}\theta_{111} = \frac{1{,}5\ \text{Å}}{2a\frac{\sqrt{3}}{3}} = \frac{1{,}5\ \text{Å}}{2(4{,}04\ \text{Å})\frac{\sqrt{3}}{3}}$$

$$\theta_{111} = \operatorname{Sen}^{-1}\left(\frac{3 \times 1{,}5\ \cancel{\text{Å}}}{2 \times (4{,}04\ \cancel{\text{Å}})\sqrt{3}}\right) \qquad \theta_{111} = 18{,}66^\circ$$

$$2\theta_{111} = 2 \times (18{,}66^\circ) = 37{,}32^\circ$$